

CUSTOMER CHURN BUSINESS ANALYSIS

Executive Summary | The Developers Arena — Week 8 Capstone

1. Business Objective

This project analyzes customer churn to identify groups associated with attrition and translate the findings into practical retention actions. The workflow covers data preparation, exploratory analysis, statistical validation, segmentation, scenario analysis, and business recommendations.

2. Key Findings

Metric	Result
Customers analyzed	500
Retained customers	447
Churned customers	53
Overall churn rate	10.60%
Month-to-month churn	20.59%
0–12 month churn	63.10%

Statistical testing supports the main findings. Monthly charges differ significantly between retained and churned customers (independent t-test, $p = 0.0101$). Contract type is significantly associated with churn (chi-square $p < 0.001$), with Cramér's $V = 0.2354$ indicating a moderate association.

3. Highest-Priority Customer Segment

The highest-priority segment is month-to-month customers with 0–12 months of tenure. It contains 35 customers, all observed as churned, producing a 100% observed churn rate in this dataset. Its average monthly charge is 120.94 and total monthly charge exposure is 4,233.00.

4. Business Recommendations

- **First-year retention:** structured onboarding and 30/60/90-day engagement checks.
- **Month-to-month customers:** suitable contract-upgrade incentives and loyalty benefits.
- **High-risk customers:** proactive, personalized retention outreach.
- **Higher monthly charges:** review pricing/value perception and plan fit.
- **Measurement:** track churn, retention, contract conversion, campaign response, and customers retained.

5. Expected Business Impact

Scenario analysis indicates that retaining 20%, 30%, or 40% of the 35 observed churned customers in the highest-risk segment would correspond to approximately 7, 10, or 14 customers retained. These are hypothetical scenarios, not forecasts.

6. Implementation Plan

Identify risk segments, launch early-tenure onboarding and month-to-month conversion initiatives, then conduct proactive high-risk outreach and controlled testing of personalized offers. Review KPIs monthly and scale strategies that demonstrate measurable retention improvement.

7. Key Limitation

The analysis uses 500 records and shows an unusually concentrated churn pattern: all observed churn occurs within the first 12 months. Findings should therefore be validated with larger, representative real-world data. Association does not imply causation.